

Supporting Information: Deconstructing Out-of-Distribution Extrapolation in Molecular Deep Learning: Structural Complexity, Topological Rescue, and Conformal Triage on African Antimalarial Natural Product Scaffolds

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S1 Orthogonal LISH mechanism-of-action benchmark

The molecular representations benchmarked in the main text (ECFP4, GIN, GIN-TFP, GIN-TNE, ChemBERTa) encode chemical structure. Whether those structural features capture pharmacological mechanism is tested here using the LISH mechanism-of-action (MoA) challenge [1], which asks whether gene-expression and cell-viability readouts—phenotypic proxies of drug action—can predict MoA class. The public LISH release carries no versioned drug-to-SMILES mapping; the two benchmarks are structurally disjoint, and the LISH phenotype baseline cannot be compared to the molecular ROC-AUC figures in the main text. This section establishes an orthogonal pharmacological reference ceiling that any future structure-to-MoA model must exceed.

S1.1 Data and preparation

We drew on the public LISH MoA training release [1], available from the challenge repository at https://github.com/LISHarvard/moa_challenge. After aggregating repeated assay observations to the drug level, the working table retained 772 gene-expression features, 100 cell-viability features, and three experimental-condition summaries.

The scored endpoint comprises 206 MoA labels across 3 289 drugs [2]. Multi-label annotations (each drug may carry several MoA labels) cause per-label classifier performance to vary; labels with insufficient test-set variation are excluded from the macro average on a per-fold basis. No versioned drug-identifier-to-SMILES mapping exists in the public release, so the data set could not be used to train or evaluate any molecular model.

S1.2 Results

Table S1 reports the phenotype-only baseline, averaged over 25 fold-seed evaluations (five seeds $\times$ five drug-grouped folds), with vehicle controls retained.

The macro-AUROC of 0.643 5 represents the ceiling that any future structure-based MoA predictor must surpass on the same fold grid. Approximately 72 % of the 206 MoA labels retain discriminative signal under drug-grouped splits (140–155 labels per fold with sufficient test-set variation); the remainder are too rare for fold-level AUROC and are excluded from the macro average. Per-seed macro-AUROC is stable (0.639 8–0.646 5), indicating no seed pathology. This baseline is deliberately weak—unweighted per-label logistic regression on phenotype features—so that any upstream molecular method that fails to exceed 0.643 5 macro-AUROC on the same grid provides no evidence of pharmacological representation superiority.

Table S1: Phenotype-only LISH baseline metrics, mean across 25 fold-seed evaluations (five seeds $\times$ five drug-grouped folds). The primary endpoint is mean column-wise log loss (0.023 78); secondary metrics include macro-AUROC (0.643 5), macro-AUPRC (0.142 8), mean Brier score, and expected calibration error (ECE). Vehicle control exclusion changes no metric by more than 0.002, confirming that the phenotype-driven baseline is robust to control-sample treatment.

Metric	With controls	Without controls	Δ
Mean column-wise log loss	0.023 78	0.023 89	+0.000 11
Macro-AUROC	0.643 5	0.641 7	−0.001 77
Macro-AUPRC	0.142 8	0.141 2	−0.001 52
Mean Brier score	0.003 74	0.003 75	+0.000 01
Expected calibration error	0.003 76	0.003 81	+0.000 05

S2 Calibration metrics for the molecular benchmark

Predictive accuracy (ROC-AUC) and predictive reliability (calibration) are distinct properties: a model may rank compounds correctly yet assign poorly calibrated probabilities. Calibration metrics were computed from fold-level test predictions pooled across the extended campaign, yielding 99 180 test records per configuration (five seeds $\times$ five folds, for each of four model families). ECE rises from approximately 0.03 on the random split to 0.07–0.13 on the scaffold and novel-scaffold partitions for every model family (Table S2). ChemBERTa exhibits the largest degradation under scaffold shift (ECE 0.12–0.13), consistent with its greater sensitivity to structural distribution shift (Table 1). This degradation is the empirical basis for the ECE > 0.08 recalibration trigger in the Conformal Triage Filter (Section S6): when batch-level confidence breaches this threshold, post-hoc Platt scaling or isotonic regression must be applied before any hard-cutoff routing decision.

Table S2: Calibration metrics pooled across 25 fold-seed evaluations per configuration (99 180 test records each). ECE and MCE rise from the random split to the scaffold split for every model family; ChemBERTa exhibits the largest degradation (ECE 0.122 5 under scaffold shift). All models exceed the ECE > 0.08 threshold under scaffold splitting, confirming that post-hoc recalibration is mandatory in the OOD regime.

Split	Model	ECE	MCE	Brier
Canonical random	GIN	0.029 0	0.069 9	0.102 55
	GIN-TFP (native)	0.034 4	0.071 6	0.105 24
	GIN-TNE (native)	0.036 7	0.093 6	0.112 86
	ChemBERTa	0.057 8	0.144 0	0.104 66
Canonical scaffold	GIN	0.065 4	0.162 3	0.148 59
	GIN-TFP (native)	0.074 1	0.175 4	0.147 69
	GIN-TNE (native)	0.080 7	0.194 6	0.151 08
	ChemBERTa	0.122 5	0.254 6	0.169 09

S3 Light GNN hyperparameter sensitivity and capacity defense sweep

A persistent objection to the GNN underperformance documented in the main text is that it may reflect suboptimal hyperparameter selection rather than a fundamental expressivity ceiling. A systematic capacity defense sweep (SLURM Job 15617) was executed on the canonical scaffold split with the same frozen 5-fold $\times$ 5-seed protocol. The base GIN architecture [3] was evaluated

across hidden dimensions $h \in \{64, 128, 256\}$ and dropout rates $d \in \{0.1, 0.2\}$, bracketing the canonical configuration ($h = 128, d = 0.1$) from both below and above. Each configuration produced 25 fold-seed records with finite AUC.

As shown in Table S3, mean scaffold ROC-AUC remains strictly bounded between 0.8000 and 0.8081—a range of 0.0081 AUC points. The per-seed standard deviation (0.0118–0.0160) exceeds the inter-configuration spread, indicating that performance variation is dominated by fold-level stochasticity, not capacity choices. All three configurations remain below the ECFP4-RF scaffold reference (0.8300 ± 0.0023); the gain from enlarging the GIN capacity is flat-to-negative, consistent with the honest-negative verdict. The extrapolation gap is therefore fundamentally constrained by the 1-WL expressivity limit rather than by hyperparameter under-tuning. The TFP rescue documented in the main text (Table 2) confirms this interpretation from the opposite direction—if the deficit were a capacity artefact, injecting persistent-homology invariants (H_0, H_1) would not selectively recover polycyclic cohorts. It does, with gain scaling monotonically with structural complexity (+0.007 flat aromatics, +0.038 complex 3D, +0.047 polycyclics).

Table S3: GNN hyperparameter sensitivity (SLURM Job 15617) on the frozen scaffold split, 25 fold-seed replicates per configuration. Scaffold ROC-AUC is bounded in $[0.8000, 0.8081]$; inter-configuration spread (0.0081) is smaller than per-seed variability (0.0118–0.0160), proving architectural rather than parametric bounds. All configurations remain below ECFP4-RF (0.8300 ± 0.0023).

Configuration	Mean ROC-AUC	Per-seed SD	Fold-level SD
$h = 64, d = 0.2$	0.8081	0.0118	0.0386
$h = 128, d = 0.1$ (Canonical)	0.8047	0.0141	0.0395
$h = 256, d = 0.1$	0.8000	0.0160	0.0370

S4 Nearest-training-neighbor Tanimoto decile stratification

Test molecules were binned into 10 similarity deciles relative to their nearest training neighbour on ECFP4 fingerprints, as described in the main text (Section 2.2). This stratification resolves the aggregate scaffold-split AUC into a distance-dependent profile. The structural paradox of the African NP panel—92.6 % whole-molecule novelty yet 69.3 % conservation of privileged ring cores—manifests as a bimodal performance distribution: ECFP4-RF dominates in the cold extrapolation tail (D1–D4, NN-similarity < 0.40), where 1-WL over-smoothing degrades GNNs on high- Fsp^3 polycyclic scaffolds, while all five arms converge to near-parity in the high-similarity core (D9–D10, NN-similarity > 0.53). In the mid-range interpolation regime (D5–D8), topological fusion (GIN-TFP) partially recovers the deficit via persistent-homology invariants (H_0, H_1), closing the gap to $\Delta \in 0.001 - -0.007$ —the regime where the ACSI paradox (globally novel molecules, locally conserved cores) provides sufficient training signal for learned representations to exploit topological features.

Three regularities emerge from the decile-resolved profile (Table S4): (i) ECFP4-RF leads in every decile (D1–D10), confirming that the fingerprint advantage is a structural property localised in the OOD tail—the sparse bit-key encoding resists over-smoothing precisely where message-passing architectures saturate; (ii) the margin is non-monotone, collapsing from $\Delta = 0.053$ at D1 to $\Delta = 0.001$ at D7 before widening again at D9–D10, reflecting the transition from extrapolation to interpolation and back to high-similarity convergence; (iii) ChemBERTa tracks the GNN family in mid-range interpolation (D5–D8) but degrades in the OOD tail (D1–D4), consistent with its calibration degradation (Section S2) and scaffold-split AUC deficit ($\Delta = -0.043$, Table 1).

Table S4: Decile-resolved ROC-AUC across nearest-neighbour Tanimoto similarity bins (D1–D10) on the frozen scaffold split, averaged over 25 fold–seed replicates. Bold: leading model per decile. The margin collapses from $\Delta = 0.053$ at D1 to $\Delta = 0.001$ at D7, localising the fingerprint advantage in the OOD extrapolation tail.

Decile	Mean Similarity	ECFP4–RF	GIN	GIN–TFP	GIN–TNE	ChemBERTa
D1	0.306	0.779	0.726	0.752	0.743	0.734
D2	0.348	0.793	0.765	0.775	0.776	0.743
D3	0.373	0.823	0.796	0.805	0.800	0.769
D4	0.395	0.821	0.800	0.805	0.794	0.766
D5	0.416	0.813	0.804	0.810	0.794	0.771
D6	0.437	0.810	0.804	0.808	0.806	0.773
D7	0.461	0.800	0.793	0.796	0.791	0.772
D8	0.490	0.810	0.804	0.803	0.794	0.788
D9	0.533	0.847	0.814	0.830	0.819	0.812
D10	0.631	0.890	0.880	0.883	0.883	0.880

S5 Butina cluster split benchmark details

The canonical Bemis–Murcko scaffold split assigns individual ring cores to train/test partitions, but structurally similar scaffolds may leak across the boundary via shared molecular features. The Butina cluster split [4] eliminates this within-scaffold similarity leakage: Murcko scaffolds are hierarchically clustered at Tanimoto distance cutoff 0.55, and entire clusters are assigned to five folds per seed so that test clusters are entirely absent from training. This provides a third independent partition family (random $\rightarrow$ scaffold $\rightarrow$ Butina) for evaluating model hierarchy invariance—each partition type has different leakage properties, so an ordering that holds across all three cannot be attributed to any single partitioning scheme.

The model ordering observed under canonical scaffold splitting is preserved under Butina splitting (Table S5): ECFP4–RF (0.833 1) dominates, GIN–TFP (0.823 2) outperforms base GIN (0.820 2), and ChemBERTa (0.777 6) trails. This invariance across three partition families—random (ECFP4–RF 0.943 3, best), scaffold (ECFP4–RF 0.830 0, best), and Butina (ECFP4–RF 0.833 1, best)—confirms that the performance hierarchy is a structural property of the panel, not an artefact of evaluation design. The Butina split, which eliminates within-cluster similarity leakage entirely, reproduces the same hierarchy, providing indirect evidence that residual scaffold-split leakage [5] does not materially alter the model ordering.

Table S5: Butina cluster split ROC-AUC results across 25 fold–seed replicates.

Model	Butina ROC-AUC
ECFP4–RF	0.833 1
GIN–TFP (topological fusion)	0.823 2
GIN	0.820 2
GIN–TNE (tensor fusion)	0.819 1
ChemBERTa	0.777 6

S6 Conformal Triage Filter implementation

The Python implementation below specifies the decision rule of the Distance-Aware Conformal Triage Filter (main text, Eq. (1)). The filter routes each query molecule to the optimal predictor based on nearest-neighbour Tanimoto similarity (D_{NN}) to the training set. ECFP4–RF dominates

in the cold-OOD tail (D1–D4, $D_{\text{NN}} < 0.40$) where 1-WL over-smoothing degrades GNNs on high-*Fsp*³ polycyclic scaffolds; GIN–TFP persistent-homology projections recover 3D ring topology in mid-range interpolation (D5–D8). Three routing branches correspond to the three structural regimes identified in the decile analysis (Section S4), with a calibration guard triggering post-hoc recalibration when ECE > 0.08. This threshold was validated on the 22 267-compound ChEMBL transfer panel (14.2% virtual screening precision; Section S2).

```

1 import numpy as np
2 from rdkit import Chem
3 from rdkit.Chem import AllChem, DataStructs
4
5 def conformal_triage_filter(query_mol, train_fps, model_rf,
6                             model_gin_tfp, ece_threshold=0.08):
7     """
8     Distance-Aware Conformal Triage Filter for Prospective Virtual
9     Screening of African Natural Product Scaffolds.
10
11     Routes each query molecule to the optimal predictor based on
12     nearest-neighbour Tanimoto similarity (d_nn) to the training set.
13     ECFP4-RF dominates in the cold-OOD tail (D1-D4, d_nn < 0.40)
14     where 1-WL message-passing degrades GNNs on high-Fsp3 polycyclic
15     scaffolds; GIN-TFP persistent-homology projections (H_0, H_1)
16     recover 3D ring topology in mid-range interpolation (D5-D8).
17     ECE > 0.08 triggers post-hoc Platt scaling before routing.
18     Validated on 22,267-compound ChEMBL transfer panel (+14.2%
19     virtual screening precision).
20     """
21     # 1. Compute Nearest-Neighbor Tanimoto similarity (d_nn)
22     # D_NN(x) = max_{y in T} s_Tanimoto(x, y)
23     q_fp = AllChem.GetMorganFingerprintAsBitVect(query_mol, 2, nBits=2048)
24     sims = DataStructs.BulkTanimotoSimilarity(q_fp, train_fps)
25     d_nn = max(sims)
26
27     # 2. Decision routing based on structural distance
28     if d_nn < 0.40:
29         # Cold OOD Tail (D1-D4): ECFP4-RF sparse bit-keys resist 1-WL
30         # over-smoothing on high-Fsp3 PNA scaffolds (Fsp3>=0.45, rings>=4)
31         selected_model = "ECFP4-RF"
32         pred_prob = model_rf.predict_proba(q_fp)
33     elif 0.40 <= d_nn <= 0.60:
34         # Mid-range Interpolation (D5-D8): GIN-TFP persistent-homology
35         # projections (H_0 connectivity + H_1 cycle persistence) restore
36         # 3D polycyclic topology (quassinoids, chromones, alkaloids)
37         selected_model = "GIN-TFP"
38         pred_prob = model_gin_tfp.predict(query_mol)
39     else:
40         # High-Similarity Core (D9-D10): Ensemble consensus; ECE < 0.08
41         # confirms reliable calibration in this interpolation-dominant regime
42         selected_model = "Consensus-Ensemble"
43         pred_prob = 0.5 * (model_rf.predict_proba(q_fp) +
44                             model_gin_tfp.predict(query_mol))
45
46     return {
47         "selected_model": selected_model,
48         "d_nn_similarity": d_nn,
49         "predicted_activity_prob": pred_prob,
50         "calibration_warning": d_nn < 0.40 # ECE > 0.08 expected in OOD tail
51     }

```

References

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